Determinants

ISC Previous Years Board Questions

With Answers

2000 to 2020

1. Using property of determinants, prove that

$$\begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3\,xyz \qquad (isc\ 2000, 1987)$$

2. Using property of determinants prove that

$$\begin{vmatrix} x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \\ xyz & xyz & xyz \end{vmatrix} = xyz(x-y)(y-z)(z-x)(xy+yz+zx) \qquad (isc\ 2001)$$

3. Using property of determinants, find the value of the following determinants :---

$$D = \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} \qquad (isc\ 2002)$$

4. Using crammer's rule, solve the following system of equations :--

$$x - y = 1,\ x + z = -6,\ x + y - 2z = 3 \qquad (isc\ \ 2002)$$

5. $If\ x \neq y \neq z\ and \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0,\ show\ that \quad xyz = -1 \qquad (isc\ 2004,1995,88)$

6. Using property of determinants show that :

$$\begin{vmatrix} a^2+1 & ab & ac \\ ba & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = 1 + a^2 + b^2 + c^2 \qquad (isc\ 2005)$$

7. $Show\ that \begin{vmatrix} 1 & 1 & 1 \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)(\alpha\beta+\beta\gamma+\gamma\alpha) \qquad (isc\ 2006)$

8. *Show that* $\begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix} = (x+y+z)^3$ $(isc\ 2007)$

9. *prove that* $\begin{vmatrix} a & b & ax+by \\ b & c & bx+cy \\ ax+by & bx+cy & 0 \end{vmatrix} = (b^2-ac)(ax^2+2bxy+cy^2)$ $(isc\ \ 2008)$

10. By using property of determinants prove that :

$\begin{vmatrix} 1+sin^2x & cos^2x & 4\ \sin 2x \\ sin^2x & 1+cos^2x & 4\ \sin 2x \\ sin^2x & cos^2x & 1+4\sin 2x \end{vmatrix} = 2+4\ \sin 2x$ $(isc\ \ 2009)$

11. By using properties of determinants, prove that the following determinant is independent of x:

$\begin{vmatrix} a & \sin x & \cos x \\ -\sin x & -a & 1 \\ \cos x & 1 & a \end{vmatrix}$ $(isc\ \ 2010)$

12. If p, q and r are not in G.P. and $\begin{vmatrix} 1 & \frac{q}{p} & \alpha+\frac{q}{p} \\ 1 & \frac{r}{q} & \alpha+\frac{r}{q} \\ p\alpha+q & q\alpha+r & 0 \end{vmatrix} = 0$,using property of determinants

show that $p\alpha^2+2q\alpha+r=0$ $(isc\ \ 2011)$

13. Prove that $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ y+z & z+x & x+y \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$ $(isc2013)$

14. Show that $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$ $(isc\ 2015)$

15. Prove that $\begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix} = 4a^2b^2c^2$ $(specimen\ paper\ 2016)$

16. Show that , using properties of determinants

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4\,abc \qquad (isc\ 2016)$$

17. Show tat $\begin{vmatrix} a & b & b+c \\ c & a & c+a \\ b & c & a+b \end{vmatrix} = (a+b+c)(a-c)^2$ $(isc\ 2017)$

18. Without expanding at any stage , find the value of

$$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} \qquad (isc\ 2018)$$

19. Using property of determinants, solve for x :

$$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0 \quad and \quad x \neq 0 \qquad (isc\ 2018)$$

20. Using determinants, find the value of k, if the area of the triangle with vertices $(-2,0), (0,4)$ and $(0,k)$ is 4 square units. $(isc\ 2019)$

21. Using properties of determinants prove that

$$\begin{vmatrix} x & x(x^2+1) & x+1 \\ y & y(y^2+1) & y+1 \\ z & z(z^2+1) & z+1 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z) \qquad (isc\ 2019)$$

22. Without expanding at any stage, find the value of the determinant :

$$\Delta = \begin{vmatrix} 20 & a & b+c \\ 20 & b & a+c \\ 20 & c & a+b \end{vmatrix} \qquad (isc\ 2020)$$

23. Using properties of determinants , show that

$$\begin{vmatrix} x & p & q \\ p & x & q \\ q & q & x \end{vmatrix} = (x-p)(x^2+px-2q^2) \qquad (isc\ \ 2020)$$

Answer : [3] $4a^2b^2c^2$ [12] $-a^3$ [18] 0 [19] $x = -(a+b+c)$ [21] 0 or 8 [22] 0

Solution of Determinants

ISC Previous Years Board Questions

2000 to 2020

1. Using property of determinants prove that :

$$\begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$$ (ISC 2000, 1987)

Solution : L.H.S.

$$= \begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix}$$

$$= \begin{vmatrix} y+z+x & x+y & x \\ z+x+y & y+z & y \\ x+y+z & z+x & z \end{vmatrix}$$ (operating $C_1 \to C_1 + C_3$)

$$= (x+y+z)\begin{vmatrix} 1 & x+y & x \\ 1 & y+z & y \\ 1 & z+x & z \end{vmatrix}$$ (taking common $(x+y+z)$ from C_1)

$$= (x+y+z)\begin{vmatrix} 1 & x+y & x \\ 0 & z-x & y-x \\ 0 & z-y & z-x \end{vmatrix}$$ (0perating $R_2 \to R_2 - R_1$ *and* $R_3 \to R_3 - R_1$)

$$= (x+y+z)\left\{1.\begin{vmatrix} z-x & y-x \\ z-y & z-x \end{vmatrix}\right\}$$ (expanding along C_1)

$$= (x+y+z)\{(z-x)^2 - (y-x)(z-y)\}$$

$$= (x+y+z)\{z^2 + x^2 - 2zx - (yz - xz - y^2 + xy)\}$$

$$= (x+y+z)\{z^2 + x^2 + y^2 - zx - yz - xy\}$$

$$= x^3 + y^3 + z^3 - 3xyz$$ Proved

2. Using property of determinants prove that :

$$\begin{vmatrix} x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \\ xyz & xyz & xyz \end{vmatrix} = xyz\,(x-y)(y-z)(z-x)(xy+yz+zx)$$ (ISC 2001)

Solution : L.H.S.

$$= \begin{vmatrix} x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \\ xyz & xyz & xyz \end{vmatrix}$$

$$= xyz \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & xz & xy \end{vmatrix} \quad \text{(taking common } x, y, z \text{ from } C_1, C_2 \text{ and } C_3 \text{ respectively)}$$

$$= xyz \begin{vmatrix} x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \\ yz & xz-yz & xy-yz \end{vmatrix} \quad \text{(operating } C_2 \to C_2 - C_1 \text{ and } C_3 \to C_3 - C_1\text{)}$$

$$= xyz \begin{vmatrix} x & y-x & z-x \\ x^2 & (y-x)(y+x) & (z-x)(z+x) \\ yz & -z(y-x) & -y(z-x) \end{vmatrix}$$

$$= xyz\,(y-x)(z-x) \begin{vmatrix} x & 1 & 1 \\ x^2 & y+x & z+x \\ yz & -z & -y \end{vmatrix}$$

$$= xyz\,(y-x)(z-x) \begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & z-y \\ yz & -z & z-y \end{vmatrix} \quad \text{(operating } C_3 \to C_3 - C_2\text{)}$$

$$= xyz\,(y-x)(z-x)(z-y) \begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & 1 \\ yz & -z & 1 \end{vmatrix}$$

$$= xyz\,(y-x)(z-x)(z-y) \begin{vmatrix} x & 1 & 0 \\ x^2-yz & y+x+z & 0 \\ yz & -z & 1 \end{vmatrix} \quad \text{(operating } R_2 \to R_2 - R_3\text{)}$$

$$= xyz\,(y-x)(z-x)(z-y) \left\{ 1. \begin{vmatrix} x & 1 \\ x^2-yz & y+x+z \end{vmatrix} \right\} \quad \text{(expanding along } C_3\text{)}$$

$$= xyz\,(y-x)(z-x)(z-y)(xy + x^2 + xz - x^2 + yz)$$

$$= xyz\,(x-y)(y-z)(z-x)(xy+yz+zx) \quad \text{Proved.}$$

3. Using property of the determinants, find the value of the following determinant:

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$$ (ISC 2002)

Solution : $\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$

$$= abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix} \quad \text{(taking common } a, b \text{ and } c \text{ from } R_1, R_2 \text{and} R_3 \text{ respectively)}$$

$$= a^2\, b^2\, c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$ (taking common a, b and c from C_1, C_2 and C_2 respectively)

$$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$$ (operating $R_2 \to R_2 + R_1$ and $R_3 \to R_3 + R_1$ respectively)

$$= a^2b^2c^2\,(-1) \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix}$$ (expanding along C_1)

$$= -\,a^2b^2c^2\,(-4)$$

$$= 4\,a^2b^2c^2$$ Answer.

4. Using Crammer's Rule , solve the following system of equations:
$x - y = 1, x + z = -6, x + y - 2z = 3$ (ISC 2002)

Solution : The given equations are $\left.\begin{array}{l} x - y = 1 \\ x + Z = -6 \\ x + y - 2z = 3 \end{array}\right\}$

Here $D = \begin{vmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & -2 \end{vmatrix} = -4$

$$D_x = \begin{vmatrix} 1 & -1 & 0 \\ -6 & 0 & 1 \\ 3 & 1 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -6 & -6 & 1 \\ 3 & 4 & -2 \end{vmatrix} = \begin{vmatrix} -6 & 1 \\ 4 & -2 \end{vmatrix} = 8$$

$$D_y = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -6 & 1 \\ 1 & 3 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & -7 & 1 \\ 1 & 2 & -2 \end{vmatrix} = \begin{vmatrix} -7 & 1 \\ 2 & -2 \end{vmatrix} = 12$$

$$D_z = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 0 & -6 \\ 1 & 1 & 3 \end{vmatrix} = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 0 & -6 \\ 2 & 1 & 4 \end{vmatrix} = -(-1) \begin{vmatrix} 1 & -6 \\ 2 & 4 \end{vmatrix} = 16$$

$\therefore\ x = \frac{D_x}{D} = \frac{8}{-4} = -2$

$y = \frac{D_y}{D} = \frac{12}{-4} = -3$

$z = \frac{D_z}{D} = \frac{16}{-4} = -4$

Hence the required solution are $x = -2, y = -3, z = -4$.

5. If $x \neq y \neq z$ and $\begin{vmatrix} x & x^2 & 1 + x^3 \\ y & y^2 & 1 + y^3 \\ z & z^2 & 1 + z^3 \end{vmatrix} = 0$, show that $xyz = -1$. (ISC 2004, 1995, 88)

Solution : Given $\begin{vmatrix} x & x^2 & 1 + x^3 \\ y & y^2 & 1 + y^3 \\ z & z^2 & 1 + z^3 \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

(taking common x, y, z $from$ R_1, R_2 and R_3 respectively)

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz\,(-1) \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix} = 0 \quad (\text{ interchanging } C_1 and C_2\ )$$

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz\,(-1)(-1) \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0 \quad (\text{interchanging } C_2 and C_3\ )$$

$$\Rightarrow (1 + xyz) \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz) \begin{vmatrix} x & x^2 & 1 \\ y - x & y^2 - x^2 & 0 \\ z - x & z^2 - x^2 & 0 \end{vmatrix} = 0 \quad (R_2 \to R_2 - R_1 \ and\ R_3 \to R_3 - R_1)$$

$$\Rightarrow (1 + xyz) \left\{ 1. \begin{vmatrix} y - x & y^2 - x^2 \\ z - x & z^2 - x^2 \end{vmatrix} \right\} = 0 \quad (\text{expanding along } C_3)$$

$$\Rightarrow (1 + xyz) \begin{vmatrix} y - x & (y - x)(y + x) \\ z - x & (z - x)(z + x) \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & y + x \\ 1 & z + x \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz)(y - x)(z - x)(z + x - y - x) = 0$$

$$\Rightarrow (1 + xyz)(y - x)(z - x)(z - y) = 0$$

$as \quad x \neq y \neq z, \quad 1 + xyz = 0 \ \ or \ \ xyz = -1$. Proved

6. Using property of determinants show that :

$$\begin{vmatrix} a^2 + 1 & ab & ac \\ ba & b^2 + 1 & bc \\ ca & cb & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$ (ISC 2005)

Solution : L.H.S. $= \begin{vmatrix} a^2 + 1 & ab & ac \\ ba & b^2 + 1 & bc \\ ca & cb & c^2 + 1 \end{vmatrix}$

$$= abc \begin{vmatrix} a+\frac{1}{a} & b & c \\ a & b+\frac{1}{b} & c \\ a & b & c+\frac{1}{c} \end{vmatrix}$$

(taking common a, b, c $from$ R_1, R_2, R_3 respectively)

$$= \begin{vmatrix} a^2+1 & b^2 & c^2 \\ a^2 & b^2+1 & c^2 \\ a^2 & b^2 & c^2+1 \end{vmatrix}$$

(multiplying C_1, C_2, C_3 by a, b, c respectively.)

$$= \begin{vmatrix} a^2+b^2+c^2+1 & b^2 & c^2 \\ a^2+b^2+c^2+1 & b^2+1 & c^2 \\ a^2+b^2+c^2+1 & b^2 & c^2+1 \end{vmatrix}$$ (operating $C_1 \to C_1 + C_2 + C_3$)

$$= (a^2+b^2+c^2+1) \begin{vmatrix} 1 & b^2 & c^2 \\ 1 & b^2+1 & c^2 \\ 1 & b^2 & c^2+1 \end{vmatrix}$$

$$= (a^2+b^2+c^2+1) \begin{vmatrix} 1 & b^2 & c^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(operating $R_2 \to R_2 - R_1 and\ R_3 \to R_3 - R_1$)

$$= (a^2+b^2+c^2+1)\left\{1.\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}\right\}$$ (expanding along C_1)

$= 1 + a^2 + b^2 + c^2$ Proved

7. Show that $\begin{vmatrix} 1 & 1 & 1 \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha\beta + \beta\gamma + \gamma\alpha)$ (ISC 2006)

Solution : L. H. S.

$$= \begin{vmatrix} 1 & 1 & 1 \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ \alpha^2 & \beta^2 - \alpha^2 & \gamma^2 - \alpha^2 \\ \alpha^3 & \beta^3 - \alpha^3 & \gamma^3 - \alpha^3 \end{vmatrix}$$ (operating $C_2 \to C_2 - C_1\ and\ C_3 \to C_3 - C_1$)

$$= \begin{vmatrix} \beta^2 - \alpha^2 & \gamma^2 - \alpha^2 \\ \beta^3 - \alpha^3 & \gamma^3 - \alpha^3 \end{vmatrix}$$ (expanding along R_1)

$$= \begin{vmatrix} (\beta+\alpha)(\beta-\alpha) & (\gamma+\alpha)(\gamma-\alpha) \\ (\beta-\alpha)(\beta^2+\alpha\beta+\alpha^2) & (\gamma-\alpha)(\gamma^2+\gamma\alpha+\alpha^2) \end{vmatrix}$$

$$= (\beta - \alpha)(\gamma - \alpha)\begin{vmatrix} \beta + \alpha & \gamma + \alpha \\ \beta^2 + \alpha\beta + \alpha^2 & \gamma^2 + \gamma\alpha + \alpha^2 \end{vmatrix}$$

$$= (\beta - \alpha)(\gamma - \alpha)\begin{vmatrix} \beta - \gamma & \gamma + \alpha \\ \beta^2 - \gamma^2 + \alpha\beta - \gamma\alpha & \gamma^2 + \gamma\alpha + \alpha^2 \end{vmatrix}$$

(operating $C_1 \to C_1 - C_2$)

$$= (\beta - \alpha)(\gamma - \alpha)\begin{vmatrix} \beta - \gamma & \gamma + \alpha \\ (\beta + \gamma)(\beta - \gamma) + \alpha(\beta - \gamma) & \gamma^2 + \gamma\alpha + \alpha^2 \end{vmatrix}$$

$$= (\beta - \alpha)(\gamma - \alpha)\begin{vmatrix} \beta - \gamma & \gamma + \alpha \\ (\beta - \gamma)(\beta + \gamma + \alpha) & \gamma^2 + \gamma\alpha + \alpha^2 \end{vmatrix}$$

$$= (\beta - \alpha)(\gamma - \alpha)(\beta - \gamma)\begin{vmatrix} 1 & \gamma + \alpha \\ \beta + \gamma + \alpha & \gamma^2 + \gamma\alpha + \alpha^2 \end{vmatrix}$$

$$= (\beta - \alpha)(\gamma - \alpha)(\beta - \gamma)\{\gamma^2 + \gamma\alpha + \alpha^2 - (\gamma + \alpha)(\beta + \gamma + \alpha)\}$$

$$= (\beta - \alpha)(\gamma - \alpha)(\beta - \gamma)\{\gamma^2 + \gamma\alpha + \alpha^2 - \gamma\beta - \gamma^2 - \gamma\alpha - \alpha\beta - \alpha\gamma - \alpha^2\}$$

$$= -(\beta - \alpha)(\gamma - \alpha)(\beta - \gamma)(\gamma\beta + \alpha\beta + \alpha\gamma)$$

$$= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha\beta + \beta\gamma + \gamma\alpha) \quad \text{Proved}$$

8. Show that $\begin{vmatrix} x - y - z & 2x & 2x \\ 2y & y - z - x & 2y \\ 2z & 2z & z - x - y \end{vmatrix} = (x + y + z)^3$ (ISC 2007)

Solution : L.H.S.

$$= \begin{vmatrix} x - y - z & 2x & 2x \\ 2y & y - z - x & 2y \\ 2z & 2z & z - x - y \end{vmatrix}$$

$$= \begin{vmatrix} x + y + z & x + y + z & x + y + z \\ 2y & y - z - x & 2y \\ 2z & 2z & z - x - y \end{vmatrix} \quad \text{(operating } R_1 \to R_1 + R_2 + R_3\text{)}$$

$$= (x + y + z)\begin{vmatrix} 1 & 1 & 1 \\ 2y & y - z - x & 2y \\ 2z & 2z & z - x - y \end{vmatrix}$$

$$= (x + y + z)\begin{vmatrix} 1 & 0 & 0 \\ 2y & -(x + y + z) & 0 \\ 2z & 0 & -(x + y + z) \end{vmatrix}$$

(operating $C_2 \to C_2 - C_1,\ C_3 \to C_3 - C_1$

$$= (x+y+z)\left\{1.\begin{vmatrix} -(x+y+z) & 0 \\ 0 & -(x+y+z) \end{vmatrix}\right\}$$

$$= (x+y+z)^3$$ Proved

9. Prove that $\begin{vmatrix} a & b & ax+by \\ b & c & bx+cy \\ ax+by & bx+cy & 0 \end{vmatrix} = (b^2-ac)(ax^2+2bxy+cy^2)$ (ISC 2008)

Solution : L.H.S. $= \begin{vmatrix} a & b & ax+by \\ b & c & bx+cy \\ ax+by & bx+cy & 0 \end{vmatrix}$

$$= \frac{1}{xy}\begin{vmatrix} ax & bx & ax^2+bxy \\ by & cy & bxy+cy^2 \\ ax+by & bx+cy & 0 \end{vmatrix}$$

(taking common $\frac{1}{x}, \frac{1}{y}$ $from\ R_1, R_2$ respectively)

$$= \frac{1}{xy}\begin{vmatrix} ax+by & bx+cy & ax^2+2bxy+cy^2 \\ by & cy & bxy+cy^2 \\ ax+by & bx+cy & 0 \end{vmatrix}$$

(operating $R_1 \to R_1 + R_2$)

$$= \frac{1}{xy}\begin{vmatrix} 0 & 0 & ax^2+2bxy+cy^2 \\ by & cy & bxy+cy^2 \\ ax+by & bx+cy & 0 \end{vmatrix}$$

(operating $R_1 \to R_1 - R_3$)

$$= \frac{1}{xy}(ax^2+2bxy+cy^2)\begin{vmatrix} by & cy \\ ax+by & bx+cy \end{vmatrix}$$ (expanding along R_1)

$$= \frac{1}{xy}(ax^2+2bxy+cy^2)\begin{vmatrix} by & cy \\ ax & bx \end{vmatrix}$$

(operating $R_2 \to R_2 - R_1$)

$$= \frac{xy}{xy}(ax^2+2bxy+cy^2)\begin{vmatrix} b & c \\ a & b \end{vmatrix}$$

$$= (ax^2+2bxy+cy^2)(b^2-ac)$$ Proved

10. By using properties of determinants, prove that : (ISC 2009)

$$\begin{vmatrix} 1+\ sin^2x & cos^2x & 4\ \sin 2x \\ sin^2x & 1+\ cos^2x & 4\ \sin 2x \\ sin^2x & cos^2x & 1+4\sin 2x \end{vmatrix} = 2+4\ \sin 2x$$

Solution : L.H.S. $= \begin{vmatrix} 1+sin^2x & cos^2x & 4\sin 2x \\ sin^2x & 1+\ cos^2x & 4\sin 2x \\ sin^2x & cos^2x & 1+4\sin 2x \end{vmatrix}$

$$= \begin{vmatrix} 1+sin^2x+cos^2x & cos^2x & 4\sin 2x \\ sin^2x+1+cos^2x & 1+cos^2x & 4\sin 2x \\ sin^2x+\ cos^2x & cos^2x & 1+4\ \sin 2x \end{vmatrix}$$

(operating $C_1 \to C_1 + C_2$)

$$= \begin{vmatrix} 2 & cos^2x & 4\sin 2x \\ 2 & 1+\ cos^2x & 4\sin 2x \\ 1 & cos^2x & 1+4\sin 2x \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -1 & 0 \\ 2 & 1+cos^2x & 4\sin 2x \\ 1 & cos^2x & 1+4\sin 2x \end{vmatrix}$$

(operating $R_1 \to R_1 - R_2$)

$$= -(-1)\begin{vmatrix} 2 & 4\sin 2x \\ 1 & 1+4\sin 2x \end{vmatrix}$$ (expanding along R_1)

$$= 2+8\sin 2x - 4\sin 2x$$

$$= 2+4\sin 2x$$ Proved

11. By using properties of determinants, prove that the following determinant is independent of x:

$$\begin{vmatrix} a & \sin x & \cos x \\ -\sin x & -a & 1 \\ \cos x & 1 & a \end{vmatrix}$$ (ISC 2010)

Solution : $\begin{vmatrix} a & \sin x & \cos x \\ -\sin x & -a & 1 \\ \cos x & 1 & a \end{vmatrix}$

$$= \begin{vmatrix} a-\ \sin x\cos x & 0 & \cos x - a\sin x \\ -\sin x + a\cos x & 0 & 1+a^2 \\ \cos x & 1 & a \end{vmatrix}$$ operating $R_1 \to R_1 - \sin x\, R_3$, $R_2 \to R_2 + aR_3$

$$= -1\begin{vmatrix} a-\sin x\cos x & \cos x - a\sin x \\ -\sin x + a\cos x & 1+a^2 \end{vmatrix}$$ (expanding along C_2)

$$= -\{(a-\sin x\cos x)(1+a^2) - (\cos x - a\sin x)(-\sin x + a\cos x)\}$$

$$= -\{(a-\sin x\cos x + a^3 - a^2\sin x\cos x) - (-\cos x\sin x + asin^2x + acos^2x - a^2\sin x\cos x)\}$$

$= -\{a + a^3 - a(sin^2x + cos^2x)\}$

$= -\{a + a^3 - a\}$

$= -a^3$ which is independent of x. Proved.

12. If p, q and r not in G.P. and $\begin{vmatrix} 1 & \frac{q}{p} & \alpha + \frac{q}{p} \\ 1 & \frac{r}{q} & \alpha + \frac{r}{q} \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$, using property of determinants, show that $p\alpha^2 + 2q\alpha + r = 0$. (ISC 2011)

Solution : Given $\begin{vmatrix} 1 & \frac{q}{p} & \alpha + \frac{q}{p} \\ 1 & \frac{r}{q} & \alpha + \frac{r}{q} \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$

$$\Rightarrow \quad \frac{1}{pq}\begin{vmatrix} p & q & p\alpha + q \\ q & r & q\alpha + r \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$$

(taking common $\frac{1}{p}, \frac{1}{q}$ $from\ R_1, R_2$ respectively)

$$\Rightarrow \quad \frac{1}{pq\alpha}\begin{vmatrix} p\alpha & q\alpha & p\alpha^2 + q\alpha \\ q & r & q\alpha + r \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$$

(taking common $\frac{1}{\alpha}$ $from\ R_1$)

$$\Rightarrow \quad \frac{1}{pq\alpha}\begin{vmatrix} p\alpha + q & q\alpha + r & p\alpha^2 + 2q\alpha + r \\ q & r & q\alpha + r \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$$ (operating $R_1 \to R_1 + R_2$)

$$\Rightarrow \quad \frac{1}{pq\alpha}\begin{vmatrix} 0 & 0 & p\alpha^2 + 2q\alpha + r \\ q & r & q\alpha + r \\ p\alpha + q & q\alpha + r & 0 \end{vmatrix} = 0$$ (operating $R_1 \to R_1 - R_3$)

$$\Rightarrow \quad \frac{1}{pq\alpha}(p\alpha^2 + 2q\alpha + r)\begin{vmatrix} q & r \\ p\alpha + q & q\alpha + r \end{vmatrix} = 0$$ (expanding along R_1)

$$\Rightarrow \quad \frac{1}{pq\alpha}(p\alpha^2 + 2q\alpha + r)(q^2\alpha + qr - pr\alpha - rq) = 0$$

$$\Rightarrow \quad \frac{1}{pq\alpha}(p\alpha^2 + 2q\alpha + r)(q^2\alpha - pr\alpha) = 0$$

$$\Rightarrow \quad \frac{\alpha}{pq\alpha}(p\alpha^2 + 2q\alpha + r)(q^2 - pr) = 0$$

$$\Rightarrow \quad (p\alpha^2 + 2q\alpha + r)\left(\frac{q^2}{pq} - \frac{pr}{pq}\right) = 0$$

$$\Rightarrow \quad (p\alpha^2 + 2q\alpha + r)\left(\frac{q}{p} - \frac{r}{q}\right) = 0$$

But $\frac{q}{p} - \frac{r}{q} \neq 0$ as p, q, r are not in G.P.

$\therefore$ $p\alpha^2 + 2q\alpha + r = 0$ Proved.

13. Prove that $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ y+z & z+x & x+y \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$ (ISC 2013)

Solution : L.H.S. $= \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ y+z & z+x & x+y \end{vmatrix}$

$$= \begin{vmatrix} x-z & y-z & z \\ x^2-z^2 & y^2-z^2 & z^2 \\ z-x & z-y & x+y \end{vmatrix}$$ (operating $C_1 \to C_1 - C_3, C_2 \to C_2 - C_3$)

$$= \begin{vmatrix} x-z & y-z & z \\ (x-z)(x+z) & (y-z)(y+z) & z^2 \\ -(x-z) & -(y-z) & x+y \end{vmatrix}$$

$$= (x-z)(y-z)\begin{vmatrix} 1 & 1 & z \\ x+z & y+z & z^2 \\ -1 & -1 & x+y \end{vmatrix}$$

$$= (x-z)(y-z)\begin{vmatrix} 0 & 0 & x+y+z \\ x+z & y+z & z^2 \\ -1 & -1 & x+y \end{vmatrix}$$

(operating $R_1 \to R_1 + R_3$)

$$= (x-z)(y-z)(x+y+z)\begin{vmatrix} x+z & y+z \\ -1 & -1 \end{vmatrix}$$ (expanding along R_1)

$$= (x-z)(y-z)(x+y+z)(-x-z+y+z)$$

$$= (x-z)(y-z)(x+y+z)(y-x)$$

$$= (x-y)(y-z)(z-x)(x+y+z)$$ Proved

14. Show that $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$ (ISC 2015)

Solution : L.H.S. $= \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+a^2+b^2) & 1-a^2-b^2 \end{vmatrix}$ (operating $\begin{matrix} C_1 \to C_1 - bC_3 \\ C_2 \to C_2 + aC_3 \end{matrix}$)

$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$

$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ 0 & -a & 1-a^2+b^2 \end{vmatrix}$ (operating $R_3 \to R_3 - bR_1$)

$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 2a \\ -a & 1-a^2+b^2 \end{vmatrix}$ (expanding along C_1)

$= (1+a^2+b^2)^2 (1-a^2+b^2+2a^2)$

$= (1+a^2+b^2)^3$ Proved

15. Prove that $\begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix} = 4a^2b^2c^2$ (specimen paper 2016)

Solution : L.H.S. $= \begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix}$

$= abc \begin{vmatrix} \frac{b^2+c^2}{a} & b & c \\ a & \frac{c^2+a^2}{b} & c \\ a & b & \frac{a^2+b^2}{c} \end{vmatrix}$ (taking common a, b, c $from$ R_1, R_2, R_3 respectively)

$= \begin{vmatrix} b^2+c^2 & b^2 & c^2 \\ a^2 & c^2+a^2 & c^2 \\ a^2 & b^2 & a^2+b^2 \end{vmatrix}$ (multiplying C_1, C_2, C_2 by a, b, c respectively)

$$= \begin{vmatrix} 0 & b^2 & c^2 \\ -2c^2 & c^2+a^2 & c^2 \\ -2b^2 & b^2 & a^2+b^2 \end{vmatrix}$$ (operating $C_1 \to C_1 - C_2 - C_3$ respectively)

$$= -2 \begin{vmatrix} 0 & b^2 & c^2 \\ c^2 & c^2+a^2 & c^2 \\ b^2 & b^2 & a^2+b^2 \end{vmatrix}$$

$$= -2 \begin{vmatrix} 0 & b^2 & c^2 \\ c^2 & a^2 & 0 \\ b^2 & 0 & a^2 \end{vmatrix}$$ (operating $C_2 \to C_2 - C_1 and\ C_3 \to C_3 - C_1$)

$$= -2 \left\{ -b^2 \begin{vmatrix} c^2 & 0 \\ b^2 & a^2 \end{vmatrix} + c^2 \begin{vmatrix} c^2 & a^2 \\ b^2 & 0 \end{vmatrix} \right\}$$ (expanding along R_1)

$$= -2\{-b^2c^2a^2 - c^2a^2b^2\}$$

$$= 4\,a^2b^2c^2$$ Proved

16. Show that , using properties of determinants

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4\,abc$$ (ISC 2016)

Solution : L.H.S. $= \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

$$= \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$ (operating $R_1 \to R_1 - R_2 - R_3$)

$$= -2 \begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$= -2 \begin{vmatrix} 0 & c & b \\ b & a & 0 \\ c & 0 & a \end{vmatrix}$$ (operating $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$)

$$= -2 \left\{ -c \begin{vmatrix} b & 0 \\ c & a \end{vmatrix} + b \begin{vmatrix} b & a \\ c & 0 \end{vmatrix} \right\}$$ (expanding along R_1)

$$= -2\{-2abc\}$$

$$= 4abc$$ Proved

17. Show that $\begin{vmatrix} a & b & b+c \\ c & a & c+a \\ b & c & a+b \end{vmatrix} = (a+b+c)(a-c)^2$ (ISC 2017)

Solution : L.H.S. $= \begin{vmatrix} a & b & b+c \\ c & a & c+a \\ b & c & a+b \end{vmatrix}$

$= \begin{vmatrix} a+c+b & b+a+c & 2(a+b+c) \\ c & a & c+a \\ b & c & a+b \end{vmatrix}$ (operating $R_1 \to R_1 + R_2 + R_3$)

$= (a+b+c) \begin{vmatrix} 1 & 1 & 2 \\ c & a & c+a \\ b & c & a+b \end{vmatrix}$

$= (a+b+c) \begin{vmatrix} 1 & 1 & 0 \\ c & a & 0 \\ b & c & a-c \end{vmatrix}$ (operating $C_3 \to C_3 - C_1 - C_2$)

$= (a+b+c)(a-c) \begin{vmatrix} 1 & 1 \\ c & a \end{vmatrix}$ (expanding along C_3)

$= (a+b+c)(a-c)^2$ Proved

18. Without expanding at any stage , find the value of

$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$ (ISC 2018)

Solution : $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$

$= \begin{vmatrix} a & b & c \\ 2x & 2y & 2z \\ x & y & z \end{vmatrix}$ (operating $R_2 \to R_2 - R_1$)

$= 2 \begin{vmatrix} a & b & c \\ x & y & z \\ x & y & z \end{vmatrix}$

$= 2 \times 0$ (as R_2 *and* R_3 are identical)

$= 0$ Answer

19. Using property of determinants, solve for x:

$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0 \text{ and } x \neq 0.$ (ISC 2018)

Solution : Given $\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$

$$\Rightarrow \quad \begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+b & a \\ x+a+b+c & b & x+c \end{vmatrix} = 0 \quad \text{(operating } C_1 \to C_1 + C_2 + C_3 \text{)}$$

$$\Rightarrow \quad (x+a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & x+b & a \\ 1 & b & x+c \end{vmatrix} = 0$$

$$\Rightarrow \quad (x+a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & x & a-c \\ 0 & 0 & x \end{vmatrix} = 0 \quad \text{(operating } \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \text{)}$$

$$\Rightarrow \quad (x+a+b+c)\begin{vmatrix} x & a-c \\ 0 & x \end{vmatrix} = 0 \quad \text{(expanding along } C_1\text{)}$$

$$\Rightarrow \quad (x+a+b+c)x^2 = 0$$

But $x \neq 0$

$\therefore \; x+a+b+c = 0$

$\therefore \; x = -(a+b+c)$ Answer

20. Using determinants, find the value of k, if the area of the triangle with vertices $(-2,0)$, $(0,4)$ and $(0,k)$ is 4 square units. (ISC 2019)

Solution : Given area of the triangle $= 4$ sq units

$$\Rightarrow \quad \frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 0 & 4 & 1 \\ 0 & k & 1 \end{vmatrix} = \pm 4$$

$$\Rightarrow \quad \frac{-2}{2}\begin{vmatrix} 4 & 1 \\ k & 1 \end{vmatrix} = \pm 4 \quad \text{(expanding along } C_1 \text{)}$$

$$\Rightarrow \quad -(4-k) = \pm 4$$

$$\Rightarrow \quad k - 4 = \pm 4$$

$\therefore \; k = 8 \; or \; 0$ Answer

21. Using properties of determinants , prove that

$$\begin{vmatrix} x & x(x^2+1) & x+1 \\ y & y(y^2+1) & y+1 \\ z & z(z^2+1) & z+1 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$ (ISC 2019)

Solution : L.H.S. $= \begin{vmatrix} x & x(x^2+1) & x+1 \\ y & y(y^2+1) & y+1 \\ z & z(z^2+1) & z+1 \end{vmatrix}$

$$= \begin{vmatrix} x & x^3+x & x+1 \\ y & y^3+y & y+1 \\ z & z^3+z & z+1 \end{vmatrix}$$

$$= \begin{vmatrix} x & x^3 & 1 \\ y & y^3 & 1 \\ z & z^3 & 1 \end{vmatrix}$$ (operating $C_2 \to C_2 - C_1, C_3 \to C_3 - C_1$)

$$= \begin{vmatrix} x & x^3 & 1 \\ y-x & y^3-x^3 & 0 \\ z-x & z^3-x^3 & 0 \end{vmatrix}$$ (operating $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$)

$$= 1 \begin{vmatrix} y-x & y^3-x^3 \\ z-x & z^3-x^3 \end{vmatrix}$$ (expanding along C_3)

$$= \begin{vmatrix} y-x & (y-x)(y^2+yx+x^2) \\ z-x & (z-x)(z^2+zx+x^2) \end{vmatrix}$$

$$= (y-x)(z-x) \begin{vmatrix} 1 & (y^2+yx+x^2) \\ 1 & (z^2+zx+x^2) \end{vmatrix}$$

$$= (y-x)(z-x)\{(z^2+zx+x^2)-(y^2+yx+x^2)\}$$

$$= (y-x)(z-x)\{z^2+zx-yx-y^2\}$$

$$= (y-x)(z-x)\{z^2-y^2+zx-yx\}$$

$$= (y-x)(z-x)\{(z+y)(z-y)+x(z-y)\}$$

$$= (y-x)(z-x)(z-y)(z+y+x)$$

$$= (x-y)(y-z)(z-x)(x+y+z)$$ Answer

22. Without expanding at any stage , find the value of determinant :

$$\Delta = \begin{vmatrix} 20 & a & b+c \\ 20 & b & a+c \\ 20 & c & a+b \end{vmatrix}$$ (ISC 2020)

Solution : Given $\Delta = \begin{vmatrix} 20 & a & b+c \\ 20 & b & a+c \\ 20 & c & a+b \end{vmatrix}$

$= 20 \begin{vmatrix} 1 & a & b+c \\ 1 & b & a+c \\ 1 & c & a+b \end{vmatrix}$ (Taking common 20 from C_1)

$= 20 \begin{vmatrix} 1 & a+b+c & b+c \\ 1 & a+b+c & a+c \\ 1 & a+b+c & a+b \end{vmatrix}$ $(C_2 \to C_2 + C_3)$

$= 20\,(a+b+c) \begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & a+c \\ 1 & 1 & a+b \end{vmatrix}$

(Taking common $(a+b+c)$ from C_2)

$= 20\,(a+b+c) \times 0$ ($\because$ C_1 *and* C_2 are identical)

$= 0$ Answer

23. Using properties of determinants , show that

$$\begin{vmatrix} x & p & q \\ p & x & q \\ q & q & x \end{vmatrix} = (x-p)(x^2+px-2q^2)$$ (ISC 2020)

Solution : L. H. S. $= \begin{vmatrix} x & p & q \\ p & x & q \\ q & q & x \end{vmatrix}$

$= \begin{vmatrix} x & p-x & q \\ p & x-p & q \\ q & 0 & x \end{vmatrix}$ (operating $C_2 \to C_2 - C_1$)

$= (x-p) \begin{vmatrix} x & -1 & q \\ p & 1 & q \\ q & 0 & x \end{vmatrix}$ (taking common $(x-p)$ from C_2)

$= (x-p) \begin{vmatrix} x & -1 & q \\ p+x & 0 & 2q \\ q & 0 & x \end{vmatrix}$ (operating $R_2 \to R_2 + R_1$)

$= (x-p).1. \begin{vmatrix} p+x & 2q \\ q & x \end{vmatrix}$ (expanding by C_2)

$= (x-p)\{(p+x)x - q.2q\}$

$= (x-p)(px + x^2 - 2\,q^2)$

$= (x-p)(x^2+px-2q^2)$ Proved